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HACK THE BOX SHERLOCK - DFIR

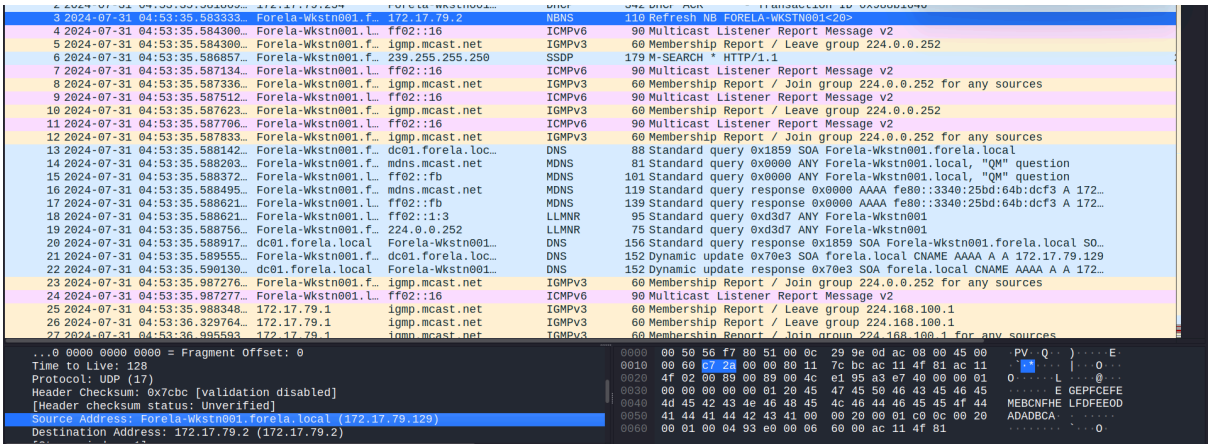
REAPER

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Reaper]
└─$ ls
Reaper  Reaper.zip

┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Reaper]
└─$ cd Reaper

┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Reaper/Reaper]
└─$ ls
ntlmrelay.pcapng  Security.evtx
```

Upon unzipping Reaper.zip, there is a PCAP file and a Windows event log (.evtx) file. Then for the better visibility and readability of the .evtx I used the Chainsaw dump tool to make it into a JSON file.

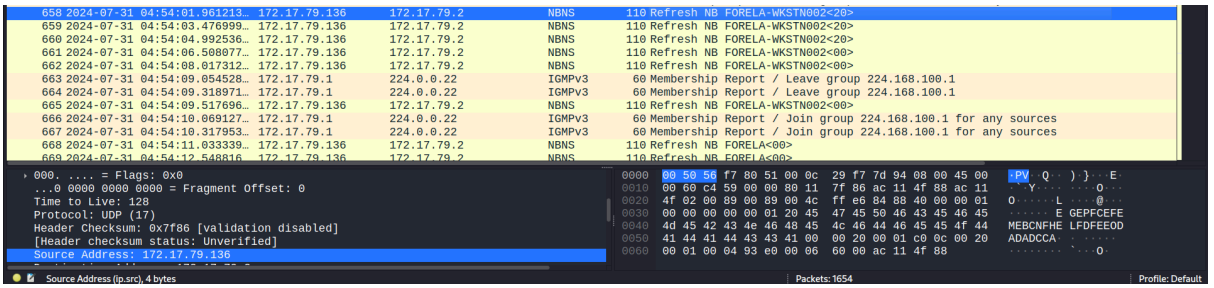
```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Reaper/Reaper]
└─$ chainsaw dump Security.evtx --json > output.json

    By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Dumping the contents of forensic artefacts from: Security.evtx (extensions: *)
[+] Loaded 1 forensic artefacts (1.1 MiB)
[+] Done
```

1. What is the IP Address for Forela-Wkstn001?



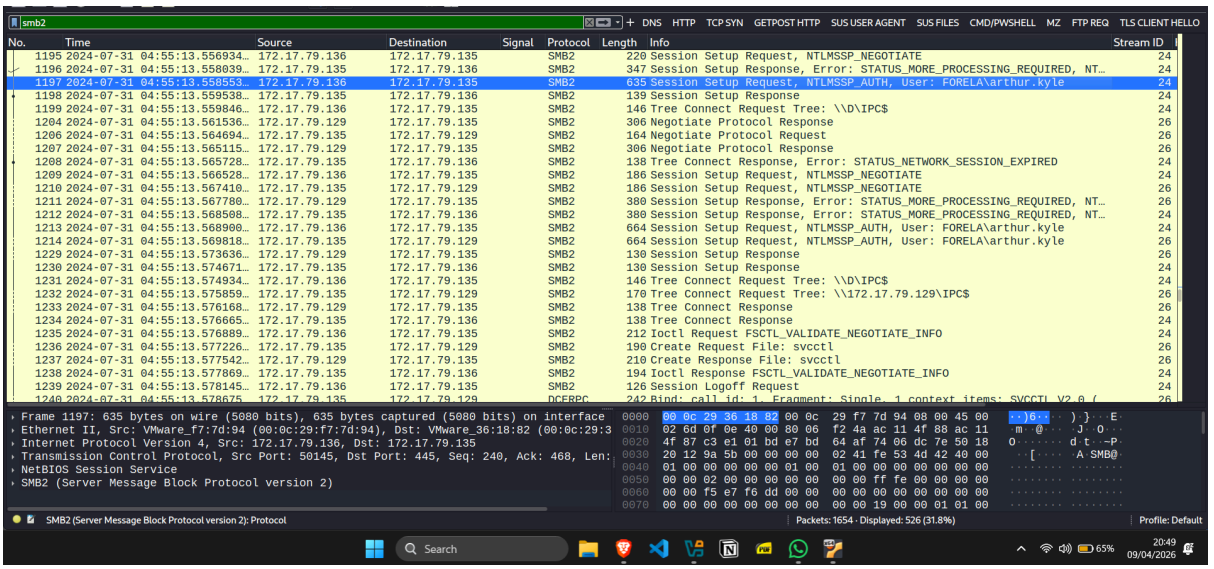
I edited in the Edit > Preferences window in Wireshark, to resolve names from IP addresses. And from packet 3, we can see in the packet details that Forela-Wkstn001's IP is **172.17.79.129**.

2. What is the IP Address for Forela-Wkstn002?



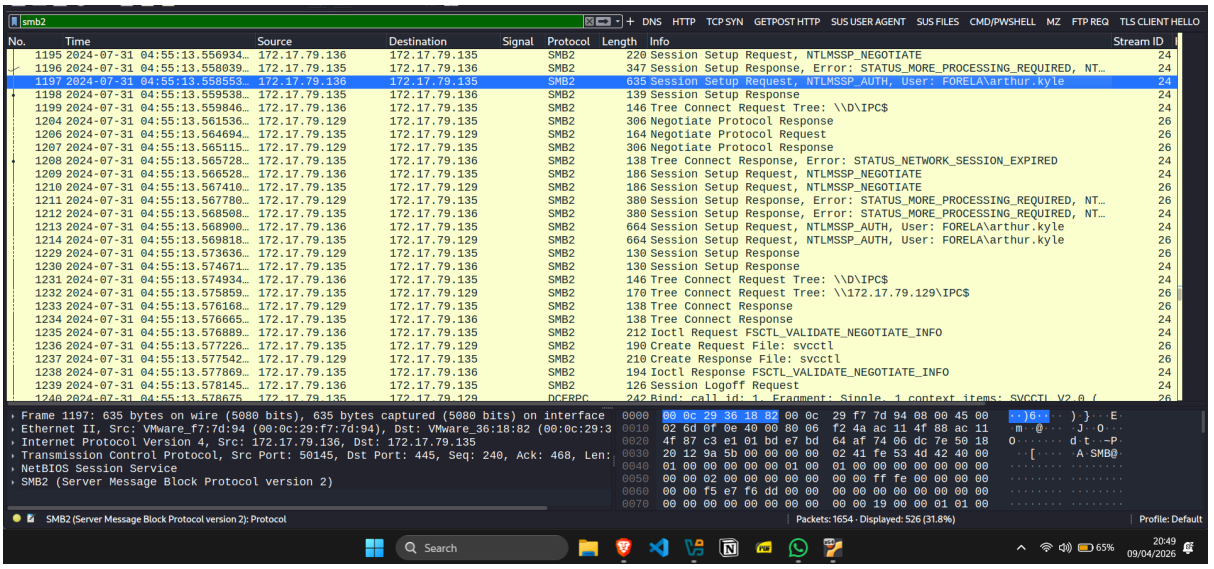
As we can see from the screenshot above, there's a packet of NBNS (NETBIOS Name Service) that is a protocol used by older Windows versions to resolve IPv4 addresses to computer names. And the answer to this question is: **172.17.79.136**.

3. What is the username of the account whose hash was stolen by attacker?



I filtered for smb2 since I thought of these as files sent with an attacker intercepting and capturing the hash. Since when users login to SMB, they send username and hashed password, this time the attacker captures that hash. And as can be seen from the screenshot above, the username is **arthur.kyle**.

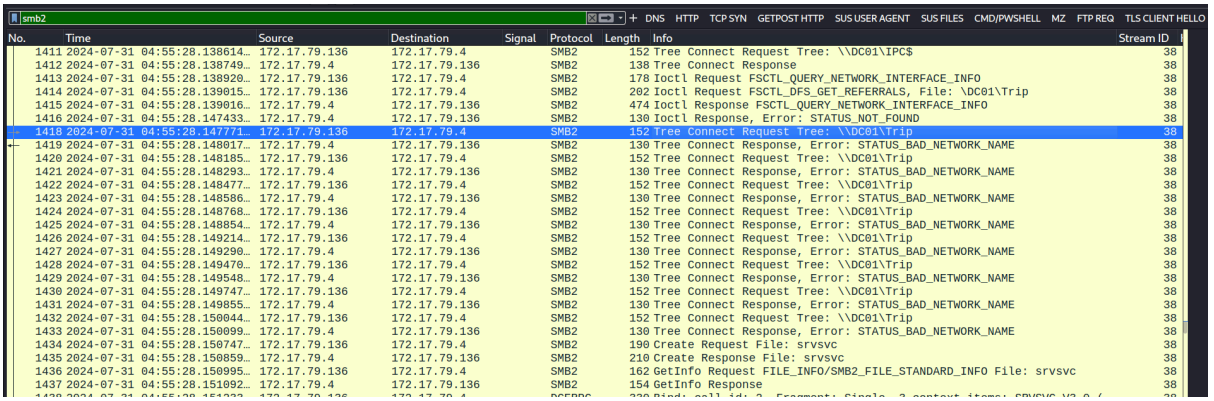
4. What is the IP Address of Unknown Device used by the attacker to intercept credentials?



No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID
1195	2024-07-31 04:55:13.556934…	172.17.79.136	172.17.79.135		SMB2	220	Session Setup Request, NTLMSSP_NEGOTIATE	24
1196	2024-07-31 04:55:13.558039…	172.17.79.135	172.17.79.136		SMB2	347	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	24
1197	2024-07-31 04:55:13.558553…	172.17.79.136	172.17.79.135		SMB2	635	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	24
1198	2024-07-31 04:55:13.559538…	172.17.79.135	172.17.79.136		SMB2	139	Session Setup Response	24
1199	2024-07-31 04:55:13.559846…	172.17.79.136	172.17.79.135		SMB2	146	Tree Connect Request Tree: \\D\IPC$	24
1204	2024-07-31 04:55:13.561536…	172.17.79.129	172.17.79.135		SMB2	306	Negotiate Protocol Response	26
1206	2024-07-31 04:55:13.564694…	172.17.79.135	172.17.79.129		SMB2	164	Negotiate Protocol Request	26
1207	2024-07-31 04:55:13.565115…	172.17.79.129	172.17.79.135		SMB2	306	Negotiate Protocol Response	26
1208	2024-07-31 04:55:13.565728…	172.17.79.135	172.17.79.136		SMB2	138	Tree Connect Response, Error: STATUS_NETWORK_SESSION_EXPIRED	24
1209	2024-07-31 04:55:13.566528…	172.17.79.136	172.17.79.135		SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE	24
1210	2024-07-31 04:55:13.567410…	172.17.79.135	172.17.79.129		SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE	26
1211	2024-07-31 04:55:13.567780…	172.17.79.129	172.17.79.135		SMB2	380	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	26
1212	2024-07-31 04:55:13.568508…	172.17.79.135	172.17.79.136		SMB2	380	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	24
1213	2024-07-31 04:55:13.568900…	172.17.79.136	172.17.79.135		SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	24
1214	2024-07-31 04:55:13.569818…	172.17.79.135	172.17.79.129		SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	26
1229	2024-07-31 04:55:13.573636…	172.17.79.129	172.17.79.135		SMB2	130	Session Setup Response	26
1230	2024-07-31 04:55:13.574671…	172.17.79.135	172.17.79.136		SMB2	130	Session Setup Response	24
1231	2024-07-31 04:55:13.574934…	172.17.79.136	172.17.79.135		SMB2	146	Tree Connect Request Tree: \\D\IPC$	24
1232	2024-07-31 04:55:13.575859…	172.17.79.135	172.17.79.129		SMB2	170	Tree Connect Request Tree: \\172.17.79.129\IPC$	26
1233	2024-07-31 04:55:13.576168…	172.17.79.129	172.17.79.135		SMB2	138	Tree Connect Response	26
1234	2024-07-31 04:55:13.576665…	172.17.79.135	172.17.79.136		SMB2	138	Tree Connect Response	24
1235	2024-07-31 04:55:13.576889…	172.17.79.136	172.17.79.135		SMB2	212	Ioctl Request FSCTL_VALIDATE_NEGOTIATE_INFO	24
1236	2024-07-31 04:55:13.577226…	172.17.79.135	172.17.79.129		SMB2	190	Create Request File: svcctl	26
1237	2024-07-31 04:55:13.577542…	172.17.79.129	172.17.79.135		SMB2	210	Create Response File: svcctl	26
1238	2024-07-31 04:55:13.577869…	172.17.79.135	172.17.79.136		SMB2	194	Ioctl Response FSCTL_VALIDATE_NEGOTIATE_INFO	24
1239	2024-07-31 04:55:13.578145…	172.17.79.136	172.17.79.135		SMB2	126	Session Logoff Request	24
1240	2024-07-31 04:55:13.578675…	172.17.79.135	172.17.79.129		DCERPC	242	Bind: call_id: 1, Fragment: Single, 1 context items: SVCCTL V2.0 (	26

Still the same screenshot as the previous question, and as have been discussed in question number 4, the traffic has been intercepted and we can see the destination IP to be **172.17.79.135**.

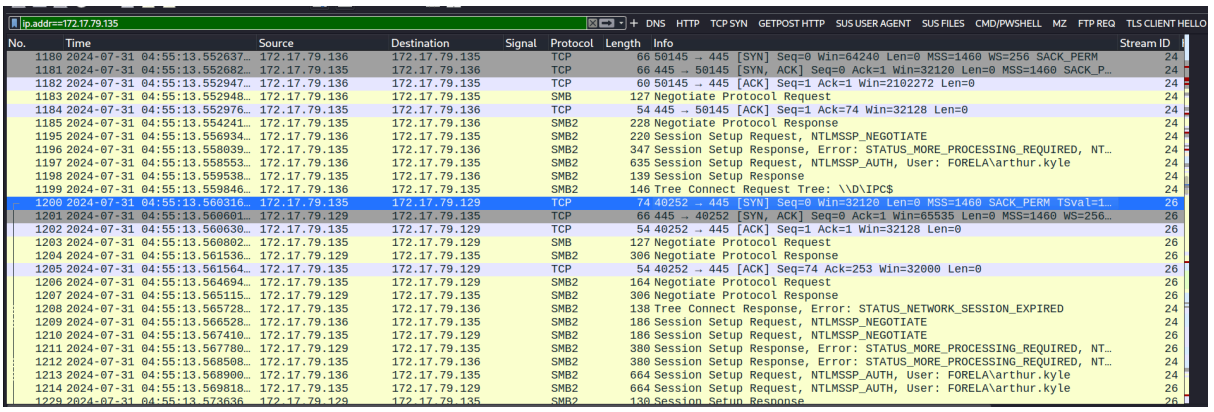
5. What was the fileshare navigated by the victim user account?



No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID
1411	2024-07-31 04:55:28.138614…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\IPC$	38
1412	2024-07-31 04:55:28.138749…	172.17.79.4	172.17.79.136		SMB2	138	Tree Connect Response	38
1413	2024-07-31 04:55:28.138920…	172.17.79.136	172.17.79.4		SMB2	178	Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO	38
1414	2024-07-31 04:55:28.139015…	172.17.79.136	172.17.79.4		SMB2	202	Ioctl Request FSCTL_DFS_GET_REFERRALS, File: \DC01\Trip	38
1415	2024-07-31 04:55:28.139016…	172.17.79.4	172.17.79.136		SMB2	474	Ioctl Response FSCTL_QUERY_NETWORK_INTERFACE_INFO	38
1416	2024-07-31 04:55:28.147433…	172.17.79.4	172.17.79.136		SMB2	130	Ioctl Response, Error: STATUS_NOT_FOUND	38
1418	2024-07-31 04:55:28.147771…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1419	2024-07-31 04:55:28.148017…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1420	2024-07-31 04:55:28.148185…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1421	2024-07-31 04:55:28.148293…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1422	2024-07-31 04:55:28.148477…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1423	2024-07-31 04:55:28.148586…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1424	2024-07-31 04:55:28.148768…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1425	2024-07-31 04:55:28.148854…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1426	2024-07-31 04:55:28.149214…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1427	2024-07-31 04:55:28.149290…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1428	2024-07-31 04:55:28.149470…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1429	2024-07-31 04:55:28.149548…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1430	2024-07-31 04:55:28.149747…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1431	2024-07-31 04:55:28.149855…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1432	2024-07-31 04:55:28.150044…	172.17.79.136	172.17.79.4		SMB2	152	Tree Connect Request Tree: \\DC01\Trip	38
1433	2024-07-31 04:55:28.150099…	172.17.79.4	172.17.79.136		SMB2	130	Tree Connect Response, Error: STATUS_BAD_NETWORK_NAME	38
1434	2024-07-31 04:55:28.150747…	172.17.79.136	172.17.79.4		SMB2	190	Create Request File: srvsvc	38
1435	2024-07-31 04:55:28.150859…	172.17.79.4	172.17.79.136		SMB2	210	Create Response File: srvsvc	38
1436	2024-07-31 04:55:28.150995…	172.17.79.136	172.17.79.4		SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc	38
1437	2024-07-31 04:55:28.151092…	172.17.79.4	172.17.79.136		SMB2	154	GetInfo Response	38

The answer is not \\DC01\IPC$ since that is not a file or directory, that is used for setup and authentication, and instead below we see another file or directory under the domain controller, and that was the answer, **\\DC01\Trip**.

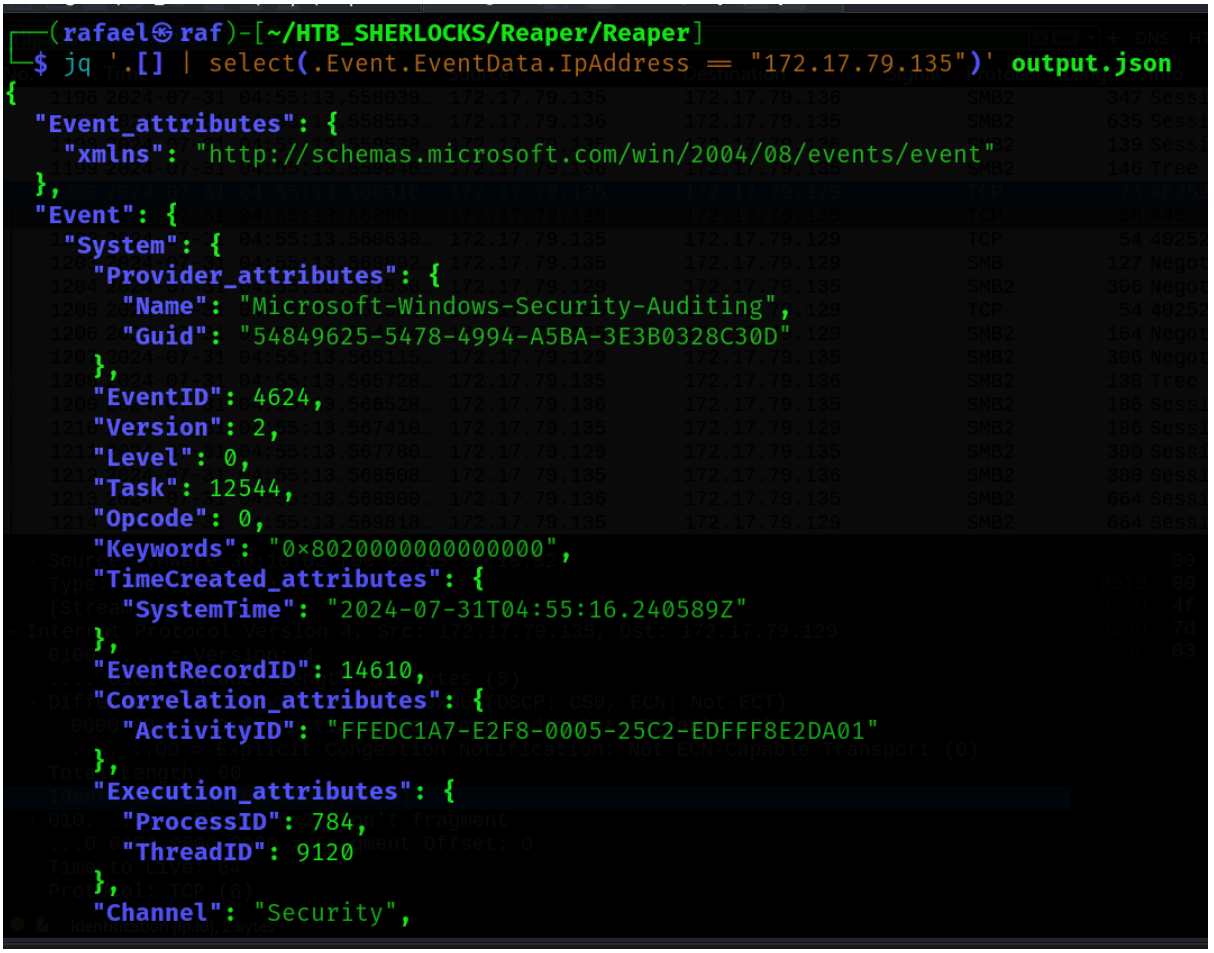
6. What is the source port used to logon to target workstation using the compromised account?



No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID
1180	2024-07-31 04:55:13.552637…	172.17.79.136	172.17.79.135		TCP	66	50145 → 445 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM	24
1181	2024-07-31 04:55:13.552682…	172.17.79.135	172.17.79.136		TCP	66	445 → 50145 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=1460 SACK_P…	24
1182	2024-07-31 04:55:13.552947…	172.17.79.136	172.17.79.135		TCP	60	50145 → 445 [ACK] Seq=1 Ack=1 Win=2102272 Len=0	24
1183	2024-07-31 04:55:13.552948…	172.17.79.136	172.17.79.135		SMB	127	Negotiate Protocol Request	24
1184	2024-07-31 04:55:13.552976…	172.17.79.135	172.17.79.136		TCP	54	445 → 50145 [ACK] Seq=1 Ack=74 Win=32128 Len=0	24
1185	2024-07-31 04:55:13.554241…	172.17.79.135	172.17.79.136		SMB2	228	Negotiate Protocol Response	24
1195	2024-07-31 04:55:13.556934…	172.17.79.136	172.17.79.135		SMB2	220	Session Setup Request, NTLMSSP_NEGOTIATE	24
1196	2024-07-31 04:55:13.558039…	172.17.79.135	172.17.79.136		SMB2	347	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	24
1197	2024-07-31 04:55:13.558553…	172.17.79.136	172.17.79.135		SMB2	635	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	24
1198	2024-07-31 04:55:13.559538…	172.17.79.135	172.17.79.136		SMB2	139	Session Setup Response	24
1199	2024-07-31 04:55:13.559846…	172.17.79.136	172.17.79.135		SMB2	146	Tree Connect Request Tree: \\D\IPC$	24
1200	2024-07-31 04:55:13.560316…	172.17.79.135	172.17.79.129		TCP	74	40252 → 445 [SYN] Seq=0 Win=32120 Len=0 MSS=1460 SACK_PERM TSval=1…	26
1201	2024-07-31 04:55:13.560601…	172.17.79.129	172.17.79.135		TCP	66	445 → 40252 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256…	26
1202	2024-07-31 04:55:13.560630…	172.17.79.135	172.17.79.129		TCP	54	40252 → 445 [ACK] Seq=1 Ack=1 Win=32128 Len=0	26
1203	2024-07-31 04:55:13.560802…	172.17.79.135	172.17.79.129		SMB	127	Negotiate Protocol Request	26
1204	2024-07-31 04:55:13.561536…	172.17.79.129	172.17.79.135		SMB2	306	Negotiate Protocol Response	26
1205	2024-07-31 04:55:13.561564…	172.17.79.135	172.17.79.129		TCP	54	40252 → 445 [ACK] Seq=74 Ack=253 Win=32000 Len=0	26
1206	2024-07-31 04:55:13.564694…	172.17.79.135	172.17.79.129		SMB2	164	Negotiate Protocol Request	26
1207	2024-07-31 04:55:13.565115…	172.17.79.129	172.17.79.135		SMB2	306	Negotiate Protocol Response	26
1208	2024-07-31 04:55:13.565728…	172.17.79.135	172.17.79.136		SMB2	138	Tree Connect Response, Error: STATUS_NETWORK_SESSION_EXPIRED	24
1209	2024-07-31 04:55:13.566528…	172.17.79.136	172.17.79.135		SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE	24
1210	2024-07-31 04:55:13.567410…	172.17.79.135	172.17.79.129		SMB2	186	Session Setup Request, NTLMSSP_NEGOTIATE	26
1211	2024-07-31 04:55:13.567780…	172.17.79.129	172.17.79.135		SMB2	380	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	26
1212	2024-07-31 04:55:13.568508…	172.17.79.135	172.17.79.136		SMB2	380	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NT…	24
1213	2024-07-31 04:55:13.568900…	172.17.79.136	172.17.79.135		SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	24
1214	2024-07-31 04:55:13.569818…	172.17.79.135	172.17.79.129		SMB2	664	Session Setup Request, NTLMSSP_AUTH, User: FORELA\arthur.kyle	26
1229	2024-07-31 04:55:13.573636…	172.17.79.129	172.17.79.135		SMB2	130	Session Setup Response	26

We can see from the highlighted packet from the screenshot above, the IP of the attacker is authenticating in using the stolen credentials, therefore the answer is **40252**.

7. What is the Logon ID for the malicious session?



```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Reaper/Reaper]
└─$ jq '.[] | select(.Event.EventData.IpAddress == "172.17.79.135")' output.json
{
  "Event_attributes": {
    "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
  },
  "Event": {
    "System": {
      "Provider_attributes": {
        "Name": "Microsoft-Windows-Security-Auditing",
        "Guid": "54849625-5478-4994-A5BA-3E3B0328C30D"
      },
      "EventID": 4624,
      "Version": 2,
      "Level": 0,
      "Task": 12544,
      "Opcode": 0,
      "Keywords": "0x8020000000000000",
      "TimeCreated_attributes": {
        "SystemTime": "2024-07-31T04:55:16.240589Z"
      },
      "EventRecordID": 14610,
      "Correlation_attributes": {
        "ActivityID": "FFEDC1A7-E2F8-0005-25C2-EDFFF8E2DA01"
      },
      "Execution_attributes": {
        "ProcessID": 784,
        "ThreadID": 9120
      },
      "Channel": "Security",
```

Here, I used the .evtx file (Windows Event Log), and I used jq to search down the Event tree all the way down to the IP address where I filtered the IP to be the attacker's IP.

```
},
"EventData": {
  "SubjectUserSid": "S-1-5-21-3239415629-1862073780-2394361899-1601",
  "SubjectUserName": "arthur.kyle",
  "SubjectDomainName": "FORELA",
  "SubjectLogonId": "0x64a799",
  "ObjectType": "File",
  "IpAddress": "172.17.79.135",
  "IpPort": "40252",
  "ShareName": "\\\\*\\IPC$",
  "ShareLocalPath": "",
  "AccessMask": "0x1",
  "AccessList": "%%4416\r\n\t\t\t\t"
}
}
}
```


And soon, as can be seen from the screenshot above, there is the LOGON ID for arthur.kyle, which is **0x64A799**.

8. The detection was based on the mismatch of hostname and the assigned IP Address.What is the workstation name and the source IP Address from which the malicious logon occur?


```
"EventData": {
  "SubjectUserSid": "S-1-0-0",
  "SubjectUserName": "-",
  "SubjectDomainName": "-",
  "SubjectLogonId": "0x0",
  "TargetUserSid": "S-1-5-21-3239415629-1862073780-2394361899-1601",
  "TargetUserName": "arthur.kyle",
  "TargetDomainName": "FORELA",
  "TargetLogonId": "0x64a799",
  "LogonType": 3,
  "LogonProcessName": "NtLmSsp ",
  "AuthenticationPackageName": "NTLM",
  "WorkstationName": "FORELA-WKSTN002",
  "LogonGuid": "00000000-0000-0000-0000-000000000000",
  "TransmittedServices": "-",
  "LmPackageName": "NTLM V2",
  "KeyLength": 128,
  "ProcessId": "0x0",
  "ProcessName": "-",
  "IpAddress": "172.17.79.135",
  "IpPort": "40252",
  "ImpersonationLevel": "%%1833",
  "RestrictedAdminMode": "-",
  "TargetOutboundUserName": "-",
  "TargetOutboundDomainName": "-",
  "VirtualAccount": "%%1843",
  "TargetLinkedLogonId": "0x0",
  "ElevatedToken": "%%1843"
}
}
```

Because this question is still related to the attacker, the answer was found under the same command tool and filter, there we find the workstation name of the attacker and the IP address that was already covered previously so the final answer would be the joined of both: **FORELA-WKSTN002, 172.17.79.135**.

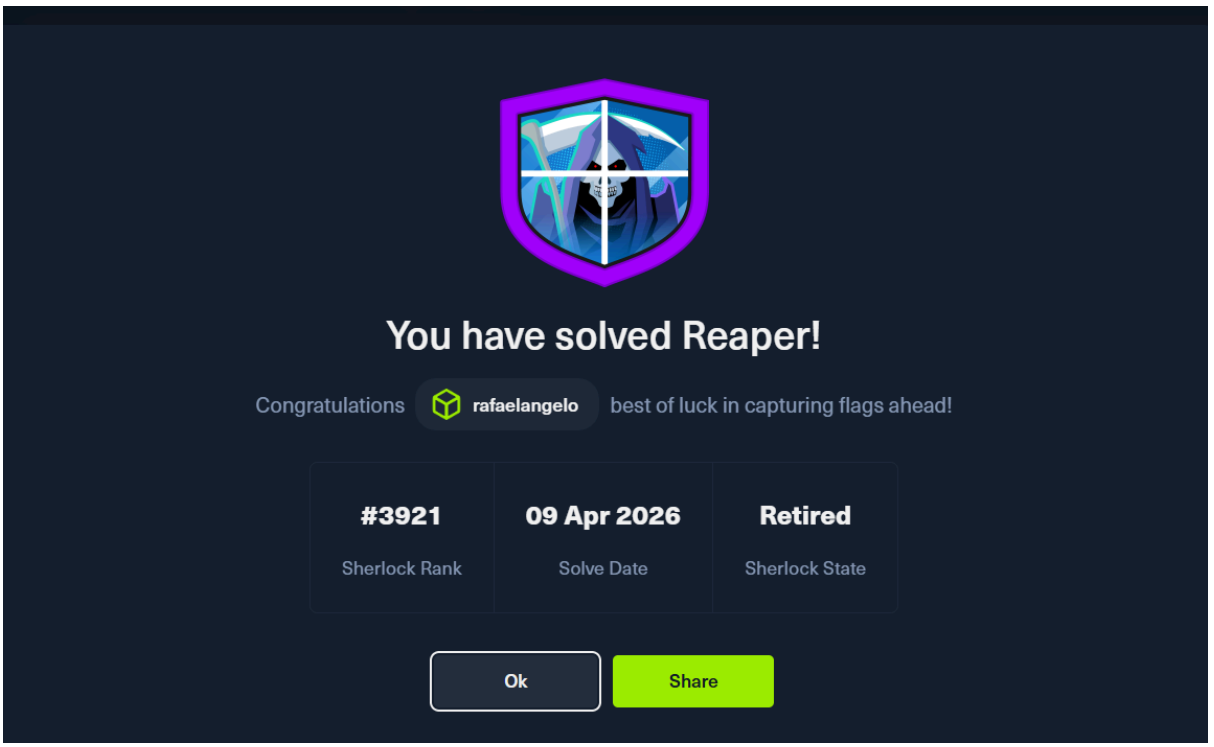
9. At what UTC time did the the malicious logon happen?


```
    "Task": 12808,
    "Opcode": 0,
    "Keywords": "0x8020000000000000",
    "TimeCreated_attributes": {
      "SystemTime": "2024-07-31T04:55:16.243325Z"
    },
    "EventRecordID": 14611,
    "Correlation": null,
    "Execution_attributes": {
      "ProcessID": 4,
      "ThreadID": 4820
    },
    "Channel": "Security",
    "Computer": "Forela-Wkstn001.forela.local",
    "Security": null
  },
  "EventData": {
    "SubjectUserSid": "S-1-5-21-3239415629-1862073780-2394361899-1601",
    "SubjectUserName": "arthur.kyle",
    "SubjectDomainName": "FORELA",
    "SubjectLogonId": "0x64a799",
    "ObjectType": "File",
    "IpAddress": "172.17.79.135",
    "IpPort": "40252",
    "ShareName": "\\\\*\\IPC$",
```


Going back to the Event of question number 7 where we got the LOGON ID, there I just got the system UTC time and that was the answer, **2024-07-31 04:55:16**.

10. What is the share Name accessed as part of the authentication process by the malicious tool used by the attacker?

```
    "Opcode": 0,
    "Keywords": "0x8020000000000000",
    "TimeCreated_attributes": {
      "SystemTime": "2024-07-31T04:55:16.243325Z"
    },
    "EventRecordID": 14611,
    "Correlation": null,
    "Execution_attributes": {
      "ProcessID": 4,
      "ThreadID": 4820
    },
    "Channel": "Security",
    "Computer": "Forela-Wkstn001.forela.local",
    "Security": null
  },
  "EventData": {
    "SubjectUserSid": "S-1-5-21-3239415629-1862073780-2394361899-1601",
    "SubjectUserName": "arthur.kyle",
    "SubjectDomainName": "FORELA",
    "SubjectLogonId": "0x64a799",
    "ObjectType": "File",
    "IpAddress": "172.17.79.135",
    "IpPort": "40252",
    "ShareName": "\\\\*\\IPC$",
    "ShareLocalPath": "",
    "AccessMask": "0x1",
    "AccessList": "%%4416\r\n\t\t\t\t"
  }
}
```


Still again the same filter and command as the last question, where the workstation name and the LOGON ID was found, highlighted in the screenshot above is the share name which was discussed earlier IPC$ is used for authentication purposes and this question number 10 is asking related to that as well, and all I had to do is reduce 1 of the "\" for each side, therefore the answer becomes ***\IPC$**.

Rafael Angelo Christianto